Assignment-2

→ One real-world application that combines both parallel computing and networked systems is distributed rendering in 3D graphics. This has been applied in sectors such as film production, game development and ~~Sti~~ scientific visualization.

How they are used:

1. Parallel computing:

i) In rendering 3D images, the process of producing high-quality images is computationally demanding. Such processes are usually parallelized to expedite the process.

ii) Each image or frame is broken down into smaller components and these smaller components are calculated simultaneously in many processing units. As an example, rendering of the lighting may be done by one processor and texture or shadows by another. By dividing these pieces across multiple processors, the time for rendering can be significantly saved.

2. Networked systems:

i) Networked systems are used by rendering farms, which are huge clusters of computers connected through a network, depending almost exclusively on networked systems to manage the task distribution.

ii) Any given machine in a rendering farm may be responsible for rendering part of a bigger image and networked systems can help manage communication between the machines in order to synchronise and assemble the ultimate image or frame.

iii) Once a node has rendered its portion of the image, it forwards the result to a central server is constructed. Networking at high speeds is essential to reduce delays in data transfer between nodes.

Why they are important?

i) Efficiency - Parallel rendering accelerates the rendering process by segmenting the task into smaller subtasks that are capable of being computed at the same time. This is especially vital in timeliness-oriented projects, like film production, where a rendering process may last for hours or even days per frame.

ii) Cost-effective Scaling - Networked systems enable the use of numerous ~~fo~~ machines in collaboration in a rendering farm and thus it becomes far more cost-effective to scale resources for rendering projects.

iii) Real-time collaboration - In shared workspaces, including studios where several artists or designers are designing different elements of the same project, networked machines facilitate real-time data transfer and communication among the team members. This improves workflow efficiency, particularly when the rendering process is outsourced to various departments.